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J.17

Ⓐ 2022-J017-1-EL-01/24

Pre-Junior Cycle Final Examination, 2022

Mathematics

Ordinary Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

EXAM ID — **Optional** 4 or 5-digit number (only if provided by your school)

NAME

SCHOOL

TEACHER

School Stamp

For Examiner only	
Question	Mark
1.	
2.	
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Total	
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Ⓐ 2022-J017-1-EL-01/24

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

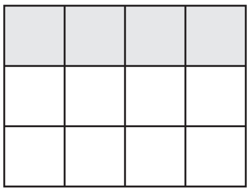
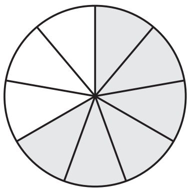
(a) Find the value of each of the following.

(i) $257 - 164$

(ii) $9 + 8{\cdot}2 \times 3$

(iii) $(313{\cdot}8 \div 6) + (72{\cdot}6 \times 3)$

(b) Two shapes, labelled **A** and **B**, are partially shaded, as shown below.
In each case, write down the fraction of the area of the shape that is shaded.
Give your answer in its simplest form.

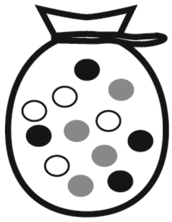
Shape A	**Shape B**
 Answer =	 Answer =

(c) In a bag there are green, red and blue counters.

$\frac{1}{4}$ of the counters are green.

$\frac{1}{3}$ of the counters are red.

What fraction of the counters are blue?



Question 2 **(Suggested maximum time: 10 minutes)**

Three year groups in a school raised money for a local charity.

2nd Year students raised €230.
3rd Year students raised €124 more than 2nd Year students.
1st Year students raised €65 less than 3rd Year students.

(a) Work out how much money 3rd Year students raised.

(b) Work out how much money 1st Year students raised.

(c) What was the total amount of money raised by **all three** year groups?

(d) Paul sold orange drinks at lunchtime at one fundraising activity.
He mixed 50 ml of orange concentrate with 200 ml of water.
Write down the ratio of orange concentrate to water. Give your answer in its simplest form.

(e) The ratio of boys to girls in a class is 2:3.
Jade says there are 27 students in the class.

(i) Explain why Jade must be wrong.

(ii) Write down the possible number of students in the class that is nearest to Jade's answer.

Question 3 **(Suggested maximum time: 5 minutes)**

(a) Jamal makes numbers using only the following digits:

4	0	1	8

He doesn't use any digit more than once in each number.

(i) Write down the largest four-digit number that Jamal can make.

(ii) Write down a three-digit number that is a multiple of 5 that Jamal can make.

(b) **(i)** Round **32 546** correct to the nearest 10. Answer =

(ii) Round **32 546** correct to the nearest 100. Answer =

(c) **(i)** List **all** the factors of 12.

(ii) List **all** the factors of 18.

(iii) Write down the **Highest Common Factor** of 12 and 18.

Question 4 **(Suggested maximum time: 5 minutes)**

The sets U, A and B are defined as follows:

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

A is the set of even numbers between 1 and 11.
B is the set of prime numbers between 1 and 10.

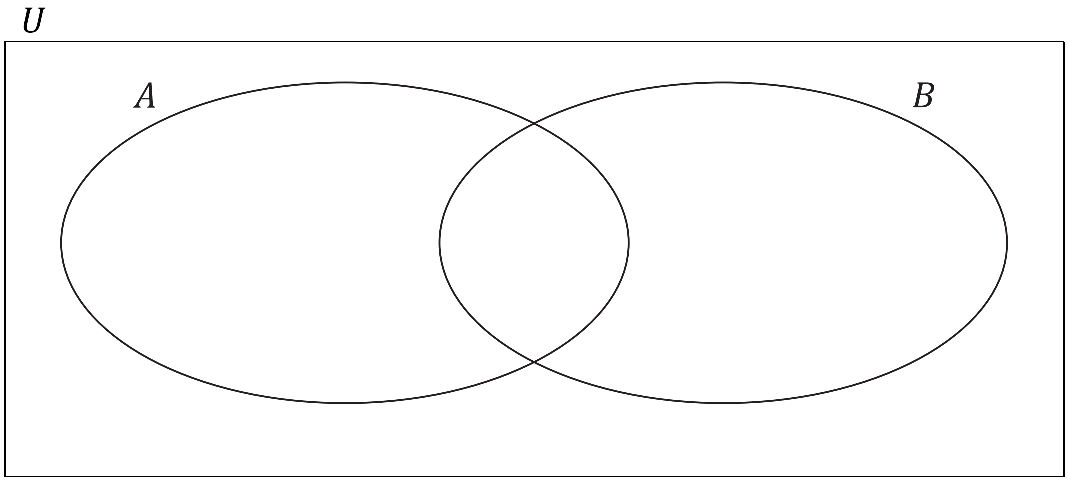
(a) List the elements of the set A.

$A = \{ \qquad \}$

(b) List the elements of the set B.

$B = \{ \qquad \}$

(c) Use the information above to fill in the following Venn diagram, placing all elements of U in the correct region.



(d) List the elements of each of the following sets:

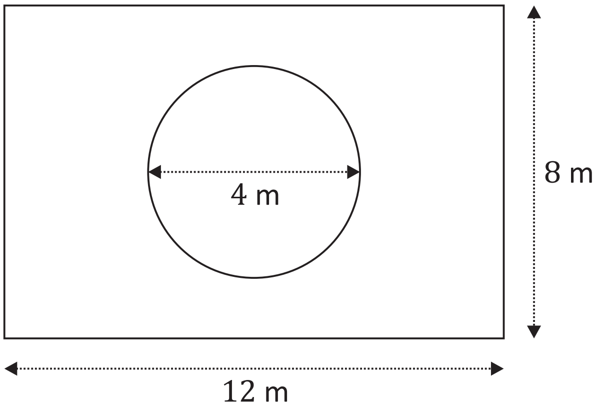
$A \cup B = \{ \qquad \}$

$A \setminus B = \{ \qquad \}$

Question 5

(Suggested maximum time: 10 minutes)

Sarah has a circular flower bed of diameter 4 m in the centre of her garden.
The garden is in the shape of a rectangle 12 m long and 8 m wide, as shown in the diagram below.
The diagram is **not** to scale.



(a) Work out the area of the garden, in m^2.

(b) Write down the radius of the flower bed.

Radius = m

(c) Work out the area of the flower bed.
Give your answer in m^2, correct to one decimal place.

(d) The remainder of the garden is made up of a grass lawn.
Work out the area of the grass lawn.
Give your answer in m^2, correct to one decimal place.

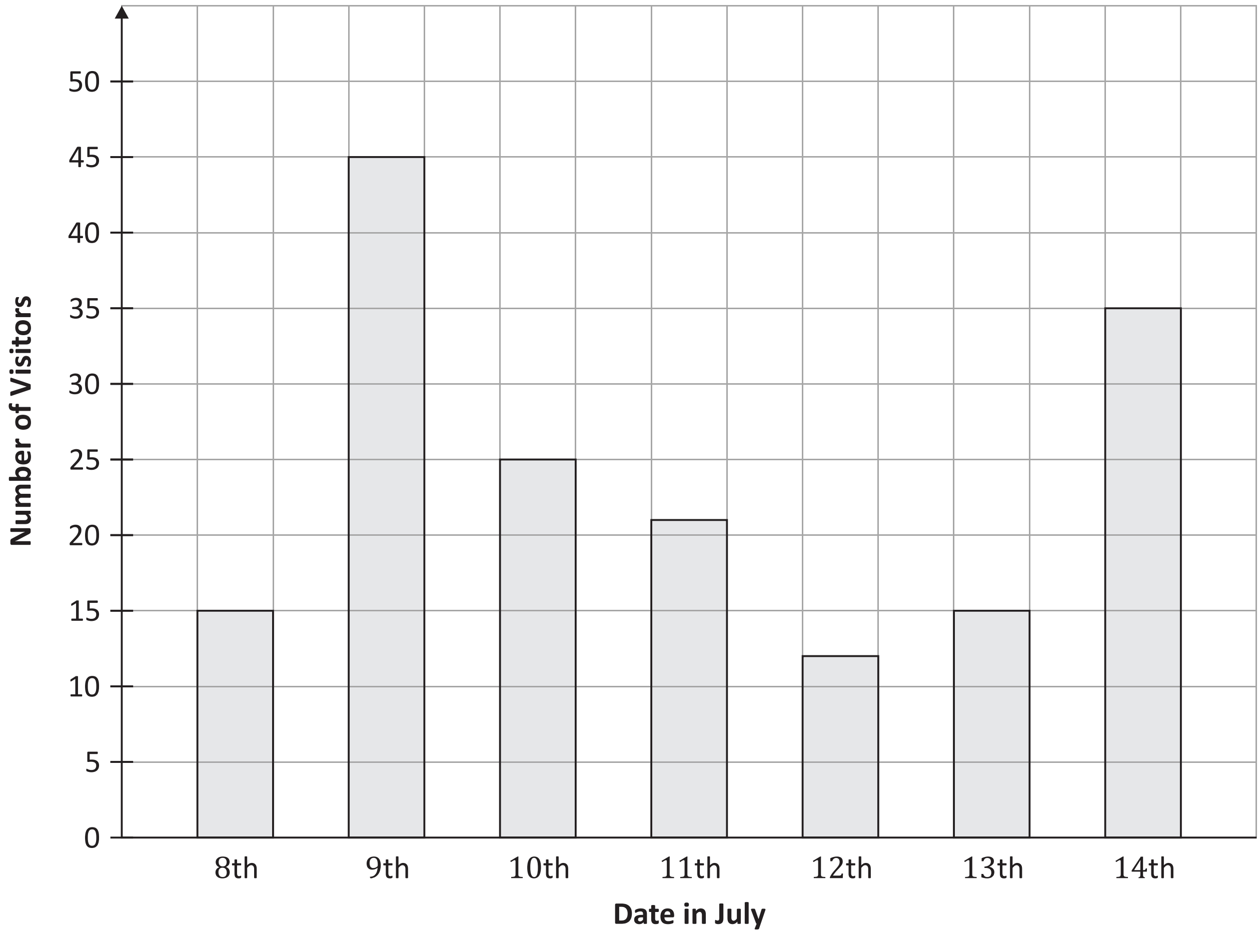
(e) Sarah wishes to plant the flower bed. She needs 14 plants to cover 1 m^2.
How many plants does she need to fill the flower bed?

(f) Sarah makes a model of her garden using the scale 1 m = 1 cm.
Construct this model in the space below, where the point P is the centre of the circle (flower bed). Show all of your construction lines clearly.

P

Question 6 **(Suggested maximum time: 15 minutes)**

The number of visitors to an art exhibition over seven days in July is shown in graph below.



(a) Use the graph to answer each of the following questions.
In each case, tick (✓) the correct box only.

(i) The **number of visitors** to the exhibition on the 12th of July was:

10	12	15	21
☐	☐	☐	☐

(ii) The date on which the **most** people visited the exhibition was:

8th July	9th July	11th July	14th July
☐	☐	☐	☐

(b) Work out the **range** of the data.

(c) Work out the **total number** of visitors to the exhibition over the seven days.

(d) Work out the **mean** number of visitors to the exhibition per day.

(e) One of visitors to the art exhibition was picked at random.
Write down the **probability** that this visitor visited the exhibition on the 11th of July.

Probability = $\frac{\square}{\square}$

(f) What **percentage** of visitors visited the exhibition on the 14th of July?
Give your answer correct to the nearest whole number.

(g) Markus correctly says 'the 8th of July was a Wednesday'.
What day of the week was the 8th of August?
Tick (✓) **one** box only. Give a reason for your answer.

Wednesday	Thursday	Friday	Saturday
☐	☐	☐	☐

Reason:

Question 7 **(Suggested maximum time: 10 minutes)**

(a) Find the value of $6x - 5y$, when $x = 4$ and $y = 3$.

(b) **(i)** Factorise the quadratic expression $x^2 - 12x + 20$. One factor is $x - 2$.

$x^2 - 12x + 20 \quad = \quad (x-2)(\qquad\qquad)$

(ii) Using your factors from part **(b)(i)**, or otherwise, solve the equation

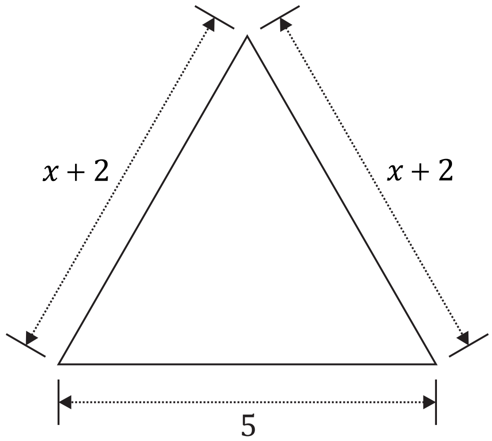
$$x^2 - 12x + 20 = 0.$$

(c) Solve the following simultaneous equations

$$3x - 2y = 11$$
$$4x + \;\; y = 11.$$

Question 8 **(Suggested maximum time: 5 minutes)**

The sides of a triangle have measurements as shown in the diagram below.



(a) Show that the **perimeter** of the triangle can be represented by the expression $2x+9$.

(b) The **perimeter** of the triangle is 15 units.
Find the value of x.

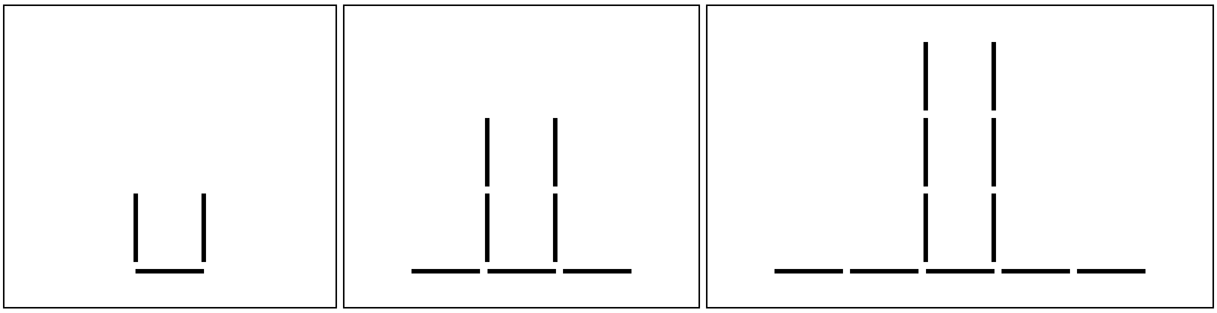
(c) What type of triangle is shown above?
Tick (✓) **one** box only. Give a reason for your answer.

isosceles ☐ equilateral ☐ right-angled ☐

Reason:

Question 9 **(Suggested maximum time: 10 minutes)**

Leo uses 1 cm sticks to make a sequence of patterns.
The first 3 patterns in his sequence are shown below.



Pattern 1	Pattern 2	Pattern 3

(a) Draw Pattern 4 in the sequence.

(b) Fill in the table below to show the number of sticks in each of the first six patterns in the sequence. One is already done for you.

Pattern	**1**	**2**	**3**	**4**	**5**	**6**
Number of sticks		7				

(c) What kind of sequence is made by the number of sticks in each pattern?
Tick (✓) **one** box only. Give a reason for your answer.

linear ☐ non-linear ☐

Reason:

(d) The number of sticks in Pattern n is:

$$4n - 1.$$

Use this to work out the number of sticks in Pattern 50 (when $n = 50$).

Question 10 **(Suggested maximum time: 5 minutes)**

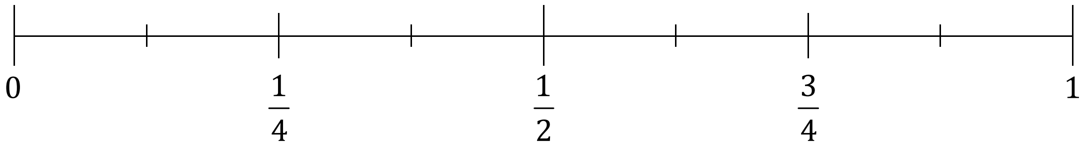
(a) The box below shows some terms used to describe the probability of an event happening.

Certain	**Unlikely**	**Impossible**	**Equal Chance**

(i) For each event **A**, **B**, **C** and **D** in the table below, use one of the terms from the box to describe the probability that it happens.

Event	Description	Probability
A	A newborn baby is a **boy**.	
B	You get a **7** when you roll a regular six-sided die.	
C	New Year's Day falls on 1st of January.	
D	It **snows** in Dublin during May.	

(ii) Write each of the letters **A**, **B**, **C** and **D** in the correct place on the probability scale below to show the probability of each event in part **(i)**.



(b) Sophia visits the school canteen with her friends. The menu is shown below.

Starter	Main Course	Dessert
• Vegetable Soup • Garlic Bread	• Fish & Chips • Shepherd's Pie • Vegetarian Lasagne	• Apple Pie • Chocolate Muffin

Sophia chooses one starter, one main course and one dessert.

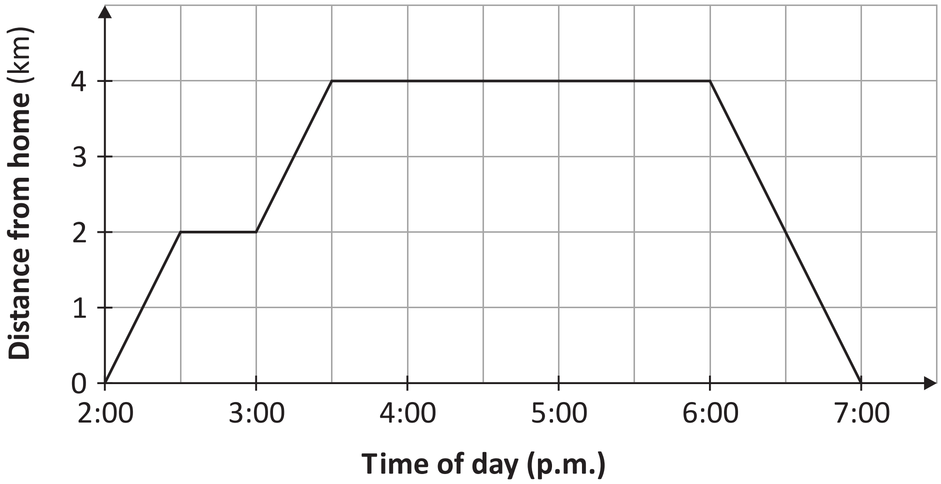
(i) Write down four different combinations that Sophia could pick.

	Starter	Main Course	Dessert
1.			
2.			
3.			
4.			

(ii) Work out the **total** number of possible lunch combinations that Sophia could choose.

Question 11 **(Suggested maximum time: 5 minutes)**

Luke walked from his home to a sports centre to meet his friends. He stopped at a shop on his way.
The graph below shows information about his journey.
For example, at 4:00 p.m., Luke had travelled 4 km from his home.



(a) Use the graph to answer each of the following questions.
In each case tick (✓) the correct box only.

(i) Luke **left home** at:

2:00 p.m.	2:30 p.m.	3:00 p.m.	3:30 p.m.
☐	☐	☐	☐

(ii) How far is the shop from Luke's home?

1 km	2 km	3 km	4 km
☐	☐	☐	☐

(iii) How long did Luke spend at the sports centre with his friends?

2 hrs	2 hrs, 30 mins	2 hrs, 45 mins	2 hrs, 50 mins
☐	☐	☐	☐

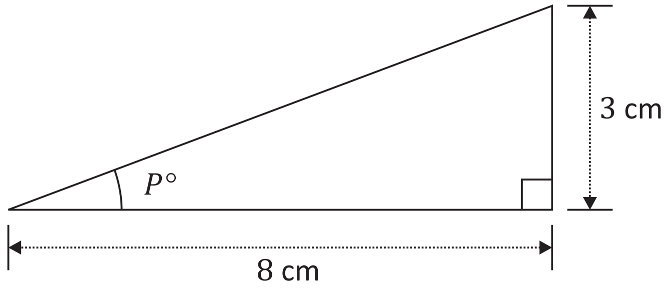
(iv) How many kilometres **in total** did Luke walk to the sports centre and home?

4 km	6 km	8 km	10 km
☐	☐	☐	☐

(b) Work out Luke's **average speed** as he walked home from the sports centre, in km per hour.

Question 12 **(Suggested maximum time: 10 minutes)**

The diagram below shows a right-angled triangle. One angle is marked P.



(a) What kind of angle is P?
Tick (✓) **one** box only.

reflex ☐ obtuse ☐ acute ☐

(b) Use the lengths of the sides of this triangle to write down the value of $\tan P$.

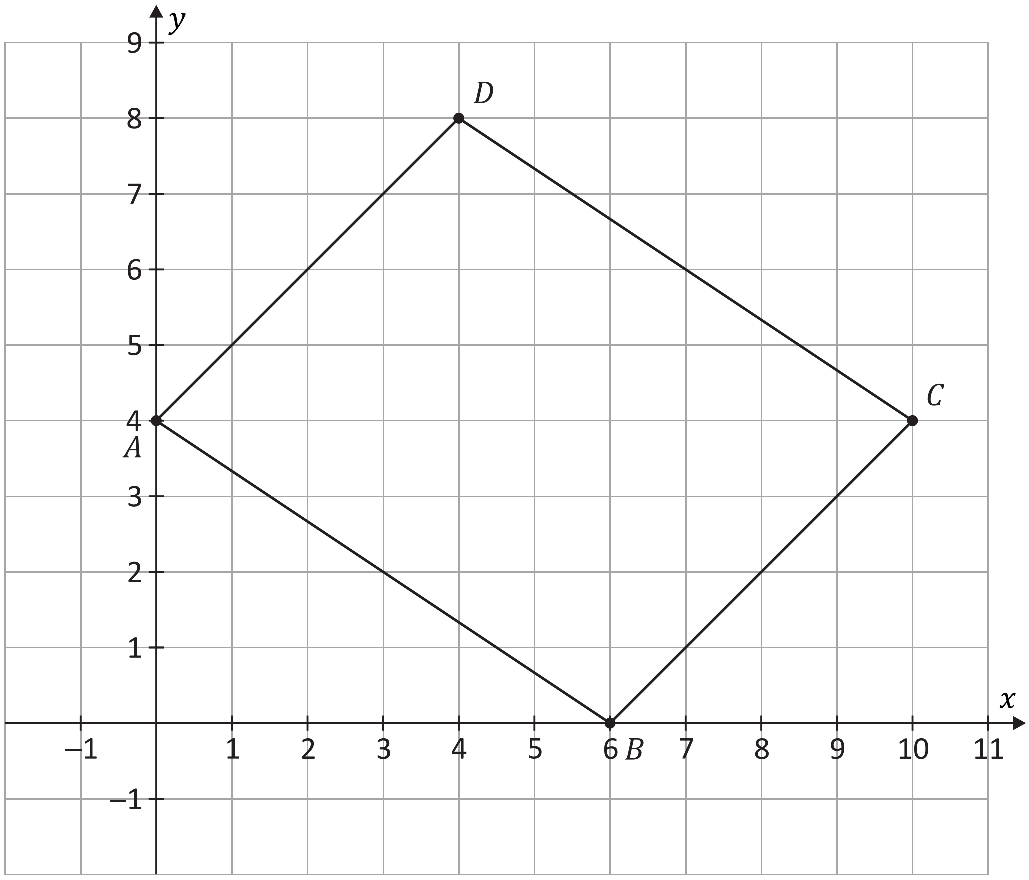
$$\tan P = \frac{\text{opposite}}{\text{adjacent}} = \frac{\square}{\square}$$

(c) Hence, use a calculator to find the **size** of the angle P, correct to the nearest degree.

(d) Peter says the length of the hypotenuse is $\sqrt{73}$ cm.
Use the theorem of **Pythagoras** to show that he is correct.

Question 13 **(Suggested maximum time: 10 minutes)**

The parallelogram $ABCD$ is shown in the co-ordinate diagram below.



(a) Complete the table below to show the co-ordinates of the four corners of $ABCD$.
One is done for you.

Point	A	B	C	D
Co-ordinates	(,)	(6, 0)	(,)	(,)

(b) Work out the co-ordinates of the **midpoint** of $[AC]$.

Midpoint of $[AC]$ = (,)

(c) Work out the **slope** of the line BD.

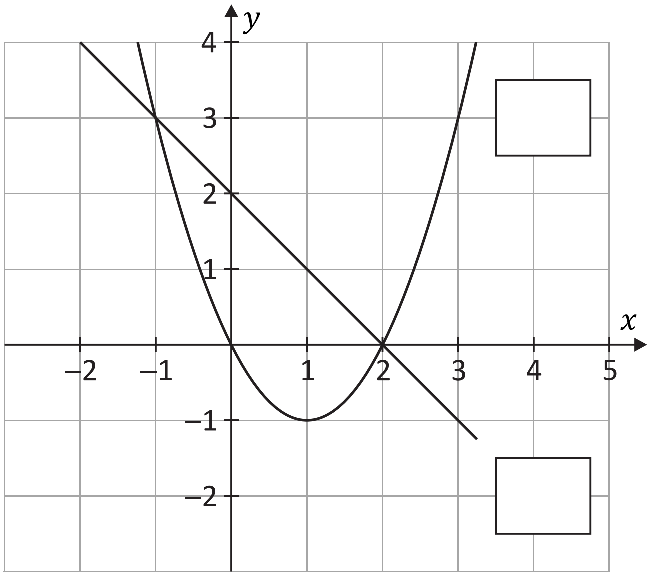
(d) Ben says that the equation of the line BD is $y = -4x + 24$.
Show that he is correct.

(e) Show that the midpoint of $[AC]$ is also on the line $y = -4x + 24$.

(f) Write down, in your own words, any theorem you know about parallelograms.

Question 14 **(Suggested maximum time: 5 minutes)**

The graphs of two functions $f(x)$ and $g(x)$ are shown on the co-ordinate diagram below.
$f(x)$ represents a straight line and $g(x)$ represents a curve.



(a) Match each of the functions to the graphs by writing $f(x)$ or $g(x)$ in the box next to the corresponding graph.

(b) Use the graphs to write down the points of intersection of $f(x)$ and $g(x)$.

Answer = (,) and Answer = (,)

Page for extra work.
Label any extra work clearly with the question number and part.

Q. & part

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Mathematics – Ordinary Level

Time: 2 hours

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